Real-Time Object Recognition and AR Overlay

Using deep learning to turn any everyday object into an augmented reality anchor

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MOTIVATION

Traditional AR needs a printed marker. What if the object itself was the marker?

PROJECT 4 GAVE US

AR rendering

3D graphics overlaid using a chessboard calibration target to estimate camera pose

PROJECT 5 GAVE US

Deep learning

CNNs and Vision Transformers trained to classify images from a camera

This project combines both. No marker required.

10 classes · 1419 images · 4-step pipeline

CLASS	IMAGES	SOURCE
book	161	Webcam
cup	173	Webcam
keyboard	152	Webcam
pen	171	Webcam
phone	159	Webcam
glasses	130	Web
headphones	110	Web
laptop	123	Web
ps5_controller	102	Web
tablet	138	Web

Webcam classes edged out web-only ones on average, though not uniformly across every class. More on this shortly.

1. **Classify**

   Crop center ROI, run MobileNetV2 to get class and confidence score

2. **Smooth and reject**

   5-frame majority-vote removes flicker; entropy check labels uncertain frames Unk

3. **Estimate pose**

   Fixed-depth pinhole model back-projects the ROI into 3D at 0.5 m

4. **Render**

   Project 8 box corners with cv2.projectPoints, draw wireframe and floating label

ARCHITECTURES AND RESULTS

Three approaches, one winner

CUSTOM CNN

ObjectCNN

- 3-layer conv network
- 64x64 RGB input
- BatchNorm + Dropout 0.4
- Trained from scratch
- 2.19M parameters

79.7% on 5 classes

VISION TRANSFORMER

ObjectViT

- 8x8 patch embedding
- 64 patches per image
- 4 transformer layers
- Trained from scratch
- 0.83M parameters

43.2% on 10 classes

TRANSFER LEARNING ❖ DEPLOYED

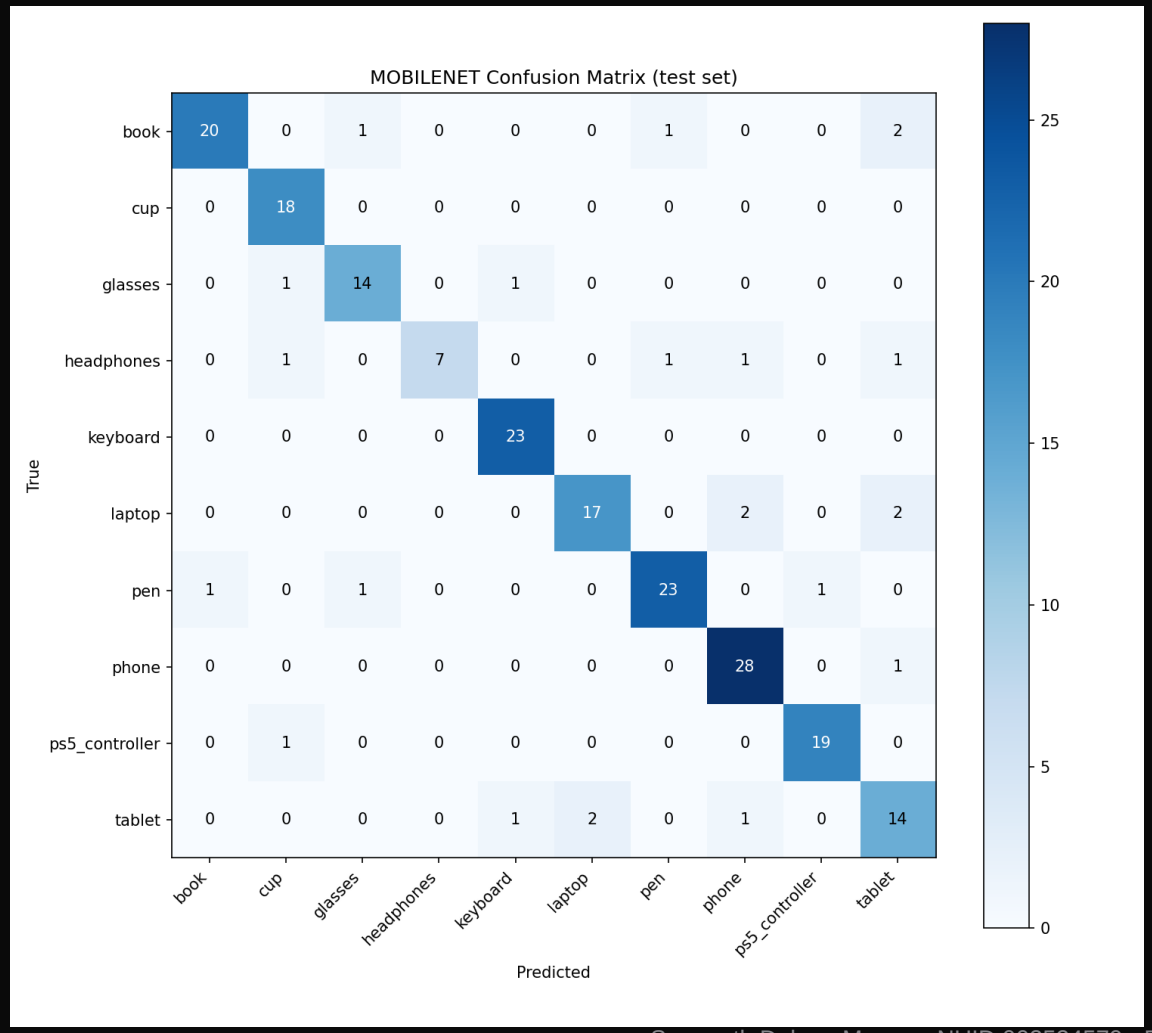
MobileNetV2

- ImageNet pretrained
- 224x224 RGB input
- Fine-tuned classifier head
- Dropout 0.3
- 2.24M parameters

88.8% on 10 classes

WHERE IT STRUGGLES

Confusion tells the story



TABLET — WEAKEST CLASS

Flat rectangular devices look nearly identical to phones and laptops at certain angles. Tablet finished at 70% precision, the lowest of all 10 classes.

HEADPHONES — CAUTIOUS MODEL

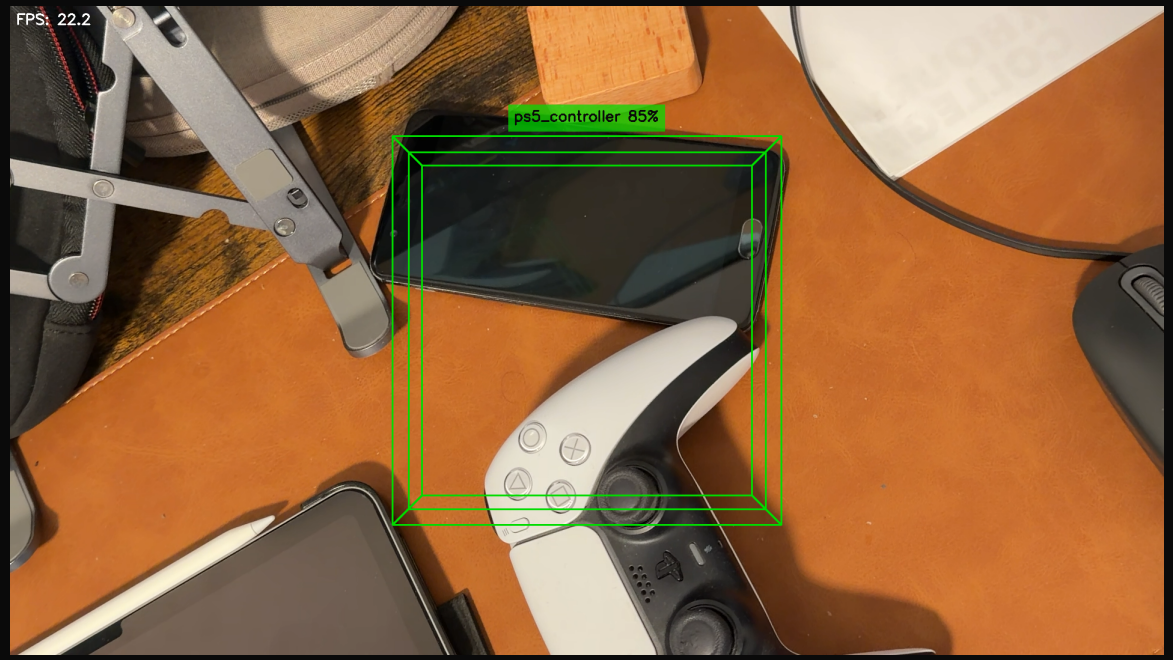
100% precision but only 63.6% recall. The model almost never guesses headphones incorrectly but often misses them entirely.

ROOT CAUSE

Errors come from visual ambiguity at class boundaries. PS5 controller (web-only) scored 95% on both metrics, so data source alone does not predict failure.

DEMO CAPTURE

AR overlay in action



A PS5 DualSense controller is in the center ROI.

ps5_controller predicted at 85% confidence.

The green 3D box is cleanly anchored with visible front and back faces. The label tag appears above the front face.

LIVE SESSION STATS

20-30

FPS live

0.50

Confidence threshold

SUMMARY

10	88.8%	20-30	0
object classes	test accuracy	fps live	printed markers

REMAINING LIMITATIONS

Fixed center ROI

Objects off to the side are not detected

Fixed depth (0.5 m)

Box scale is incorrect at very different distances

ALREADY ADDRESSED

5-frame majority-vote buffer eliminates label flickering

Entropy-based rejection shows Unknown when no familiar object is in vi